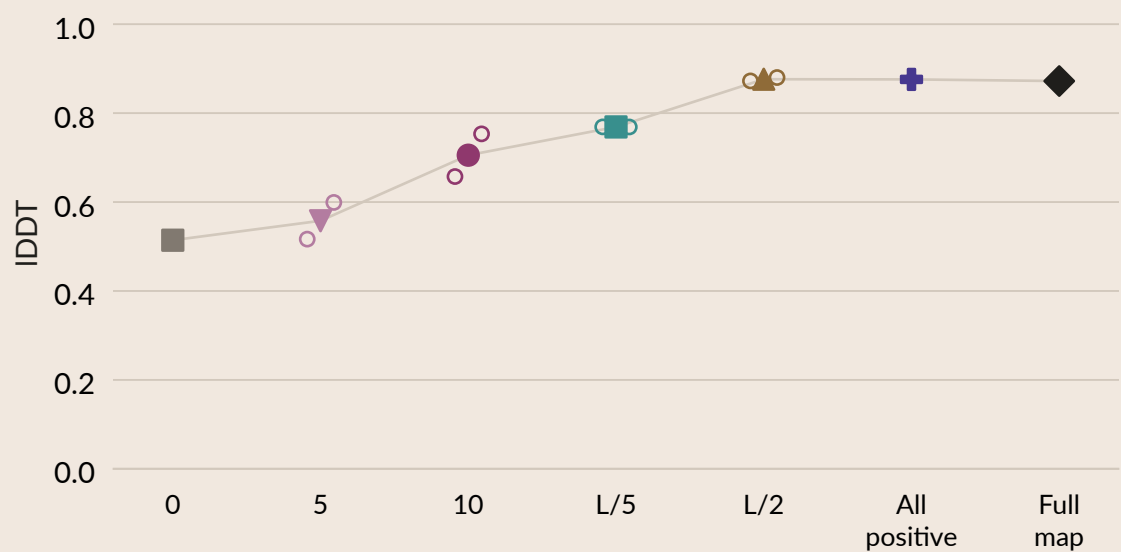


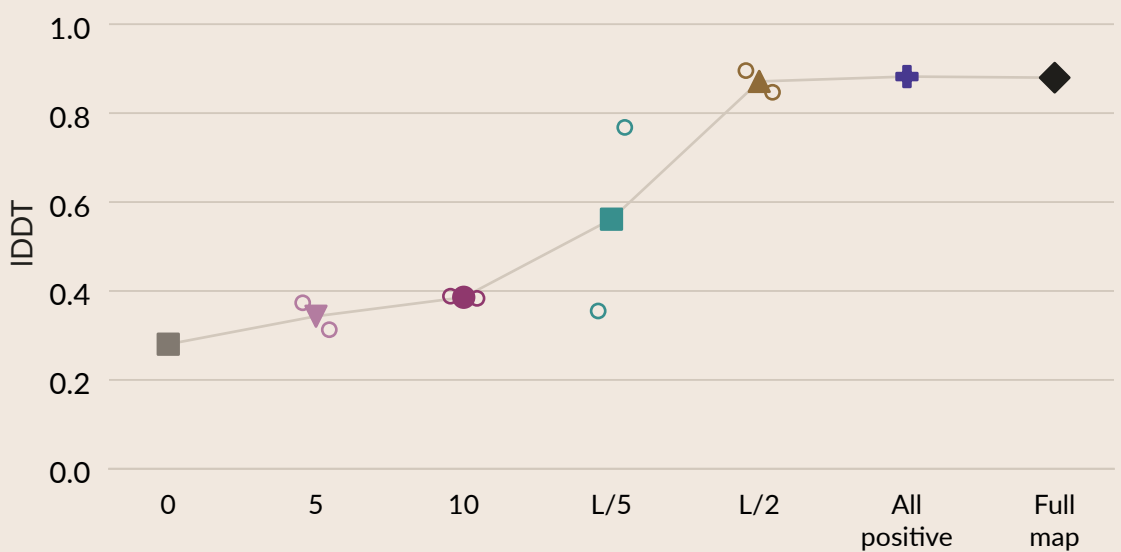
How many true contacts does Helico need at low MSA depth?

Five natural proteins • MSA depth <10 • Helico sees no MSA

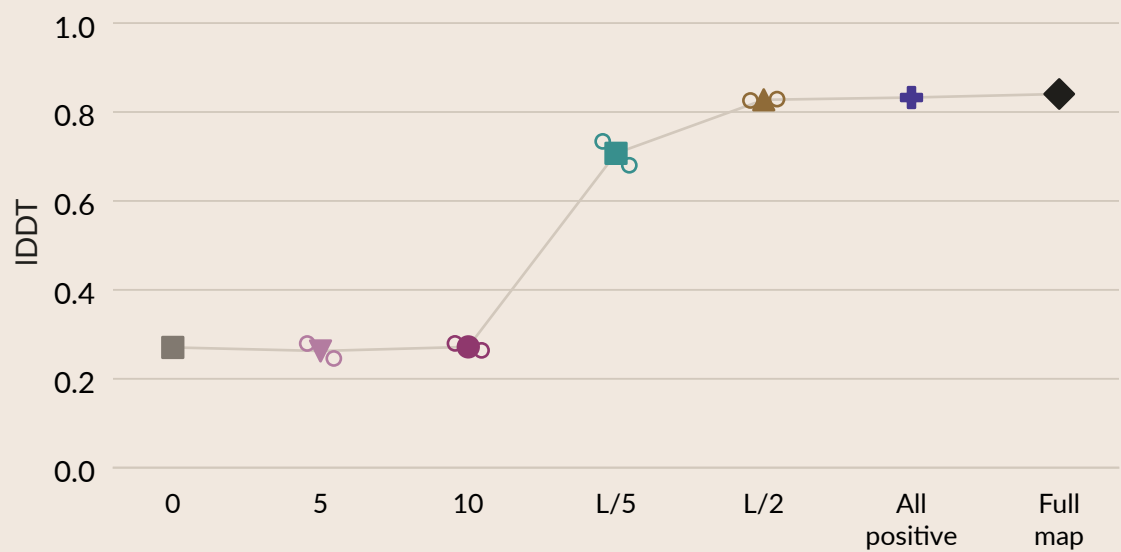
8ii8_A • depth 3 • L = 226



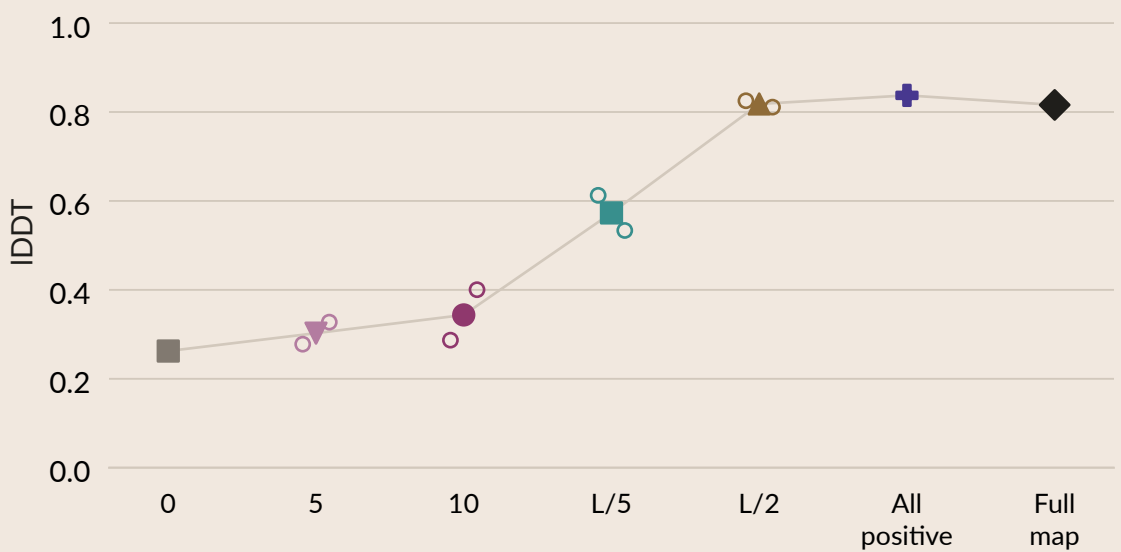
8oxk_A • depth 6 • L = 99



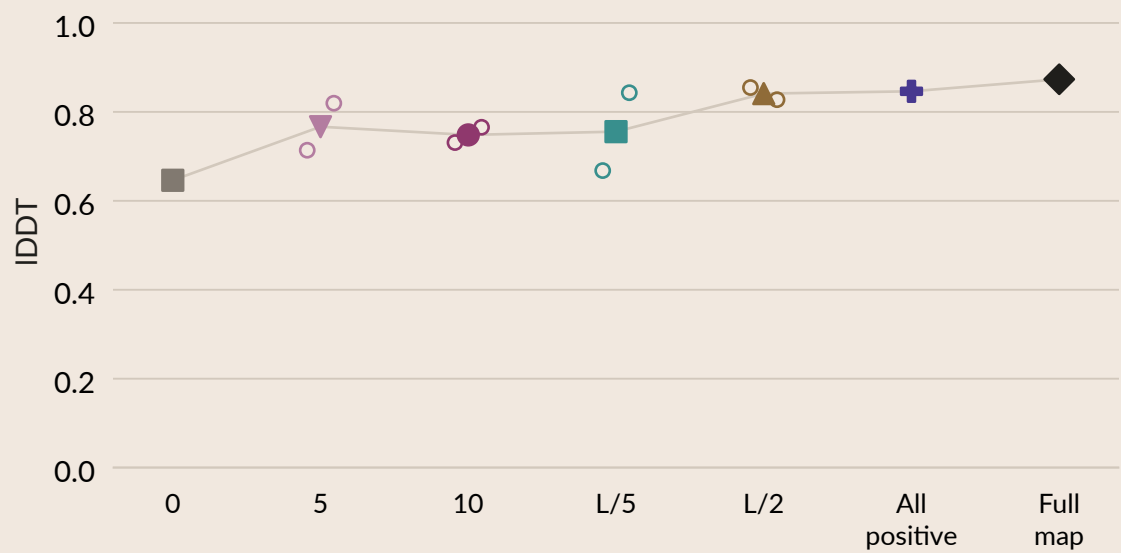
8qoh_A • depth 3 • L = 278



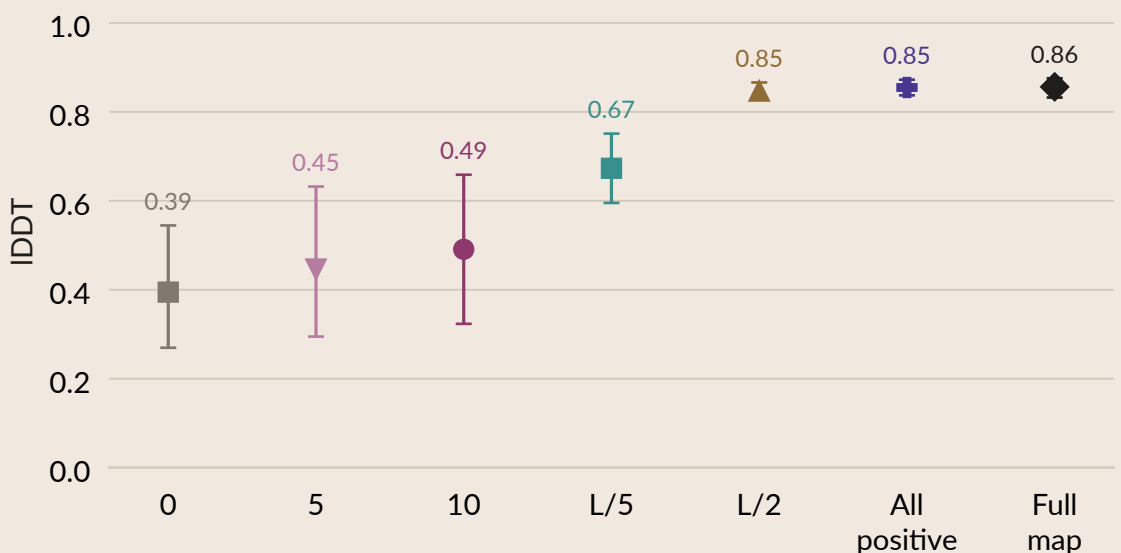
8ux2_A • depth 6 • L = 246



8wrx_A • depth 4 • L = 88



Mean of five proteins • 95% protein interval



True contacts supplied → Sparse arms: two random subsets; hollow dots are the individual subset results.

Unselected pairs stay unknown. Full map also supplies non-contacts. All contact arms use ground truth.